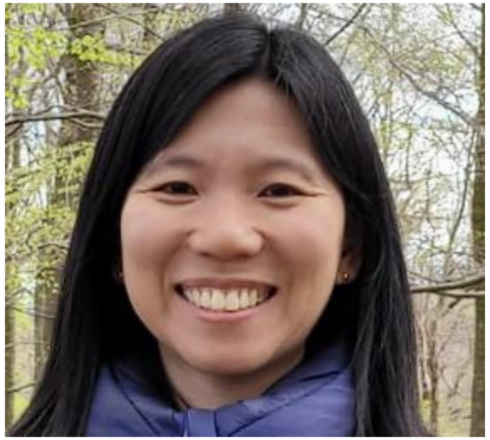
Applied Machine Learning SP4 2025

Selpi

Computer Science and Engineering

Chalmers and GU

Who will run the course?


Selpi, Examiner

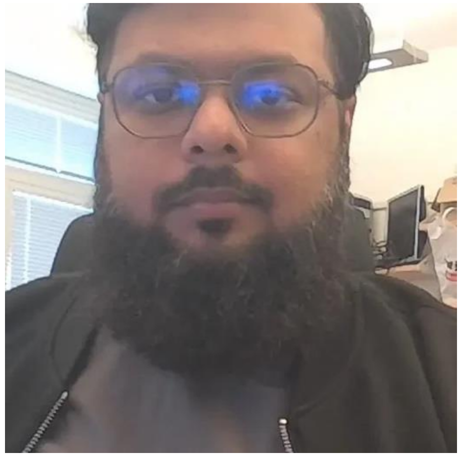

Amer, TA


Kelsey, TA


Linus, TA


Mehrdad, TA

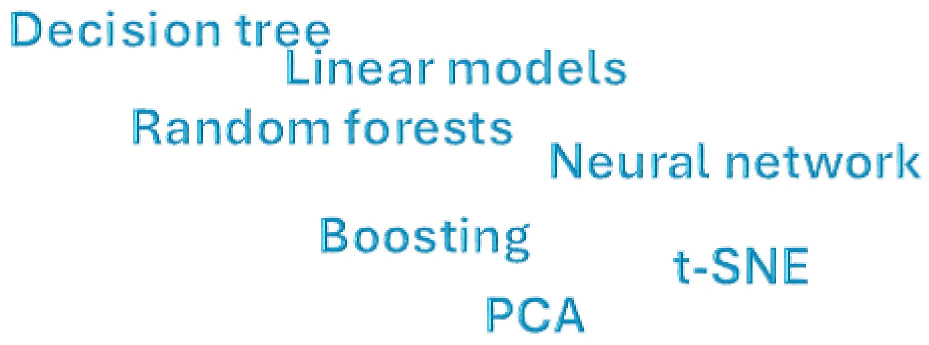

Mengyu, TA


Muhammad, TA

Stefano, TA

What topics will be covered and why?

- ML methods
  - the idea behind them
  - how they are implemented at high level
  - some use cases of them
  - how to apply them in practise



- Real-world context
  - Practical assignments through programming assignments
  - How ML is used in industry through invited talk from industry
  - Real task of annotating data
  - Ethical, legal, and interpretability issues

Why study applied machine learning (ML)?

Who are studying applied ML?

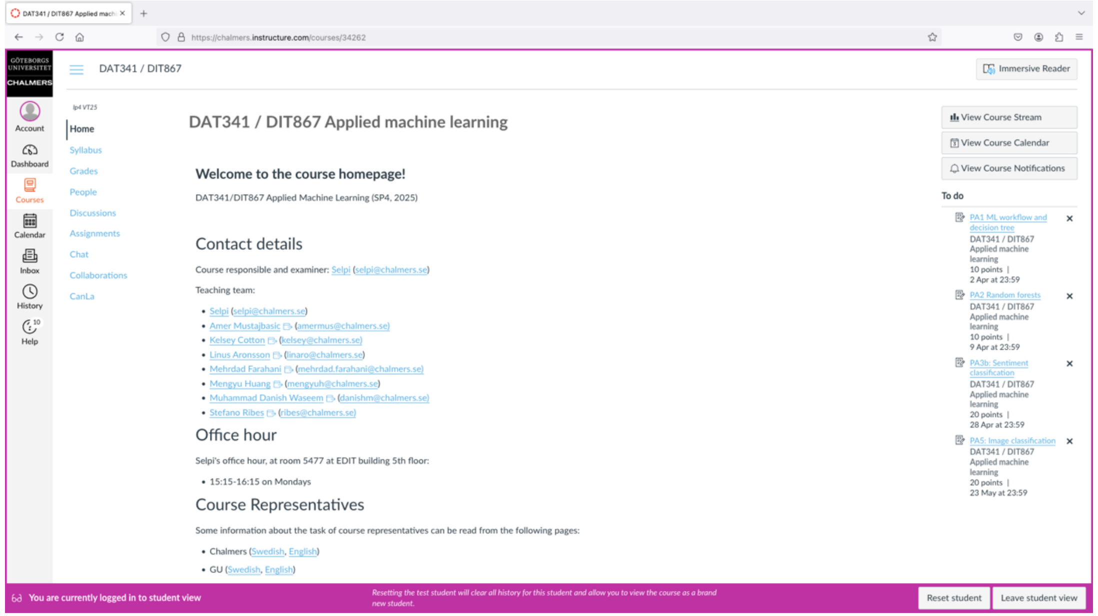
https://forms.office.com/e/FbTxnJwNFi

Survey on background of Applied ML students in SP4 2025

How will this course be run?

- [Course Canvas page]
  - Syllabus
  - Class sessions (can be lectures, flipped classrooms, discussions)
  - Preparation (can be reading, watching, studying/trying codes)

- Assessment
  - Assignments:
    - 5 programming assignments (done in group of 2)
    - 1 annotation task (done individually)
  - Take-home exam (2-3June 2025)



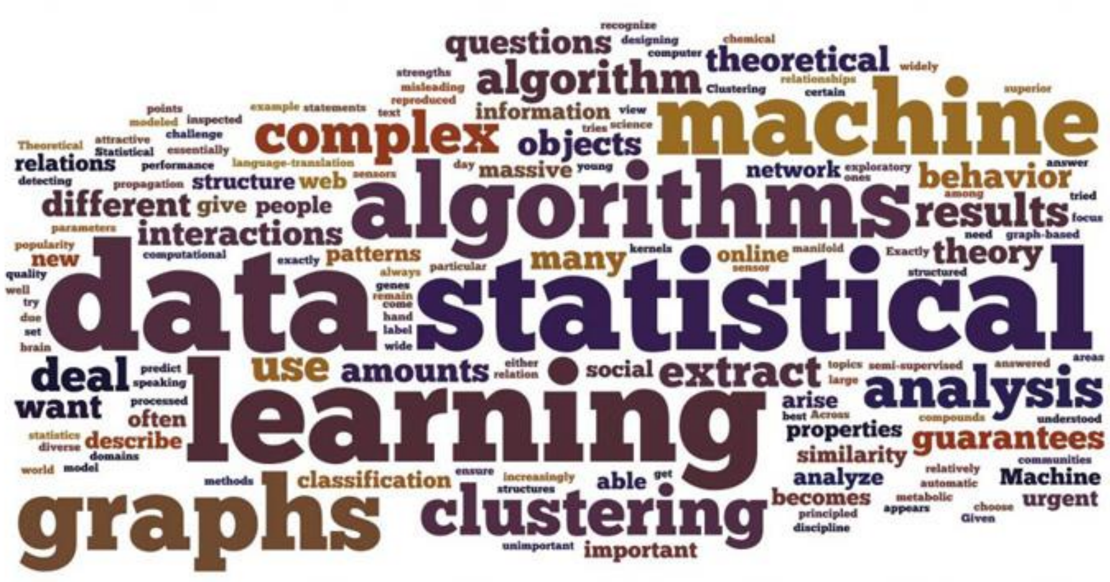
When do we meet outside class sessions?

- Mondays – office hour
  - 13:00-14:00 on 5 & 12 May
  - 15:15-16:15 for other weeks
  - Place: Selpi’s room 5477 at EDIT building
- Thursdays – physical lab
  - 08:00-09:45
- Fridays - physical lab
  - 15:15-17:00

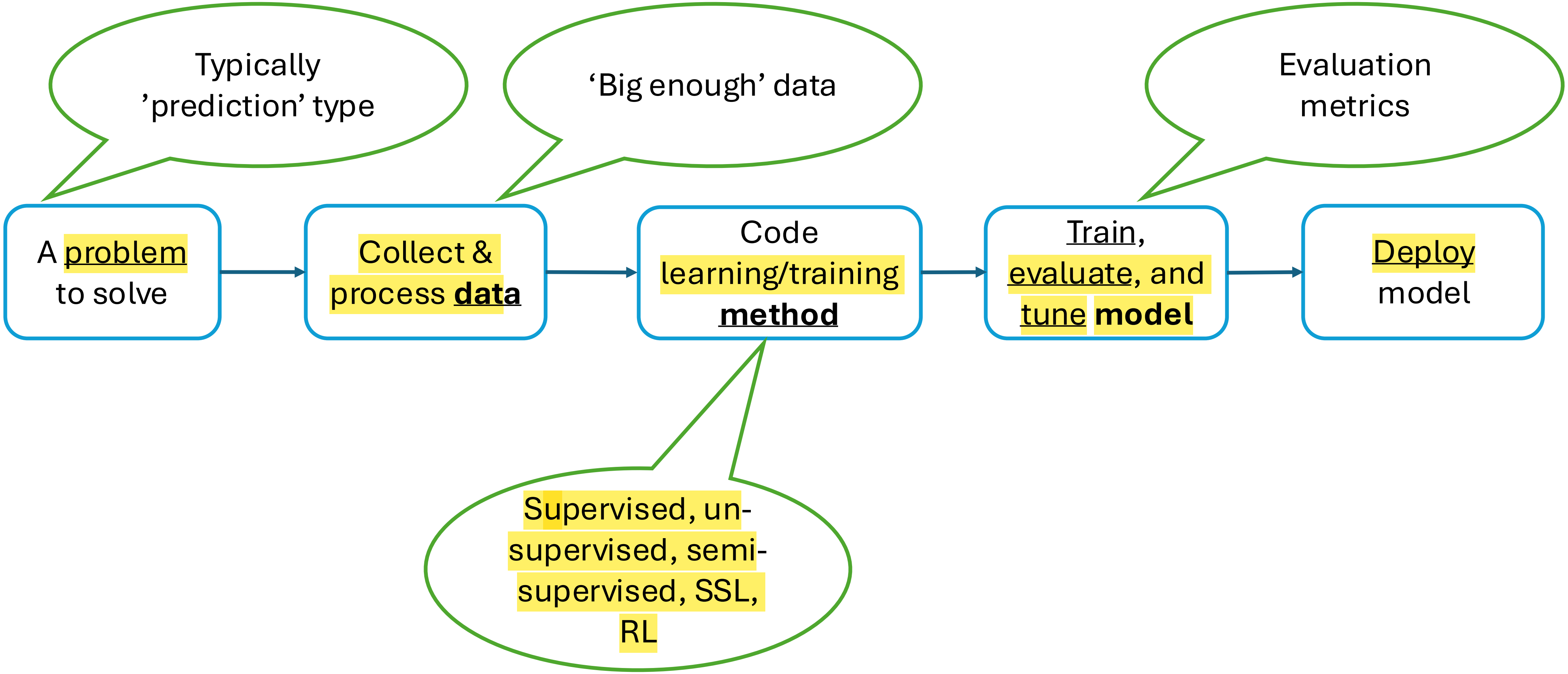
ML workflow

What is machine learning?

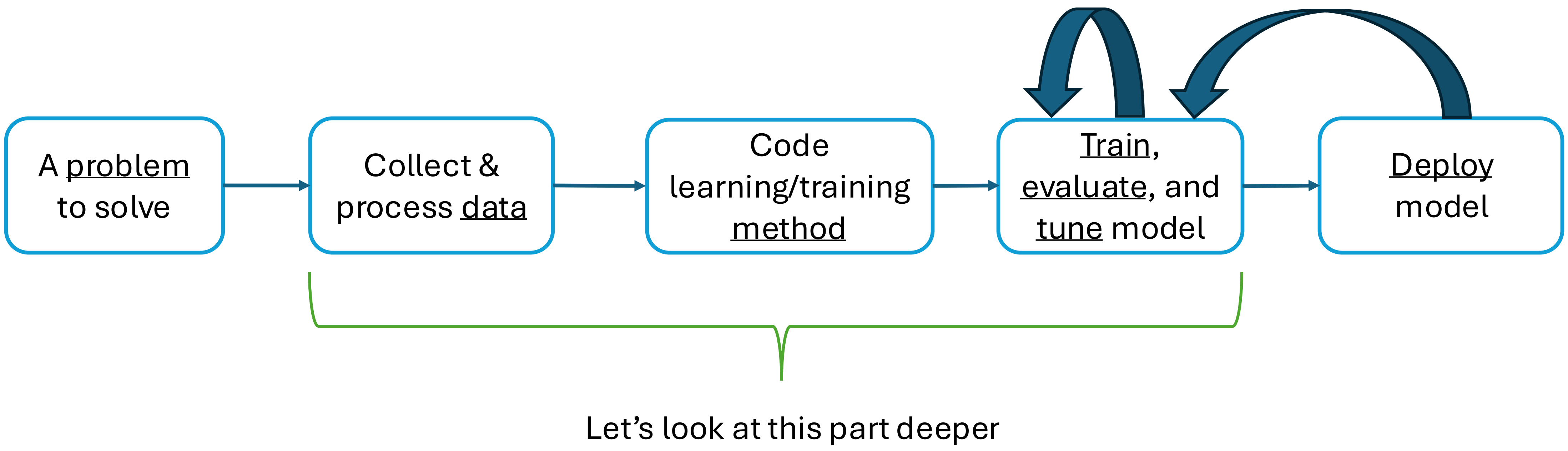
- The goal: to create a system that can learn/generalise by itself from data.
- One technique/method within AI.
- Relies on statistics or other theories.



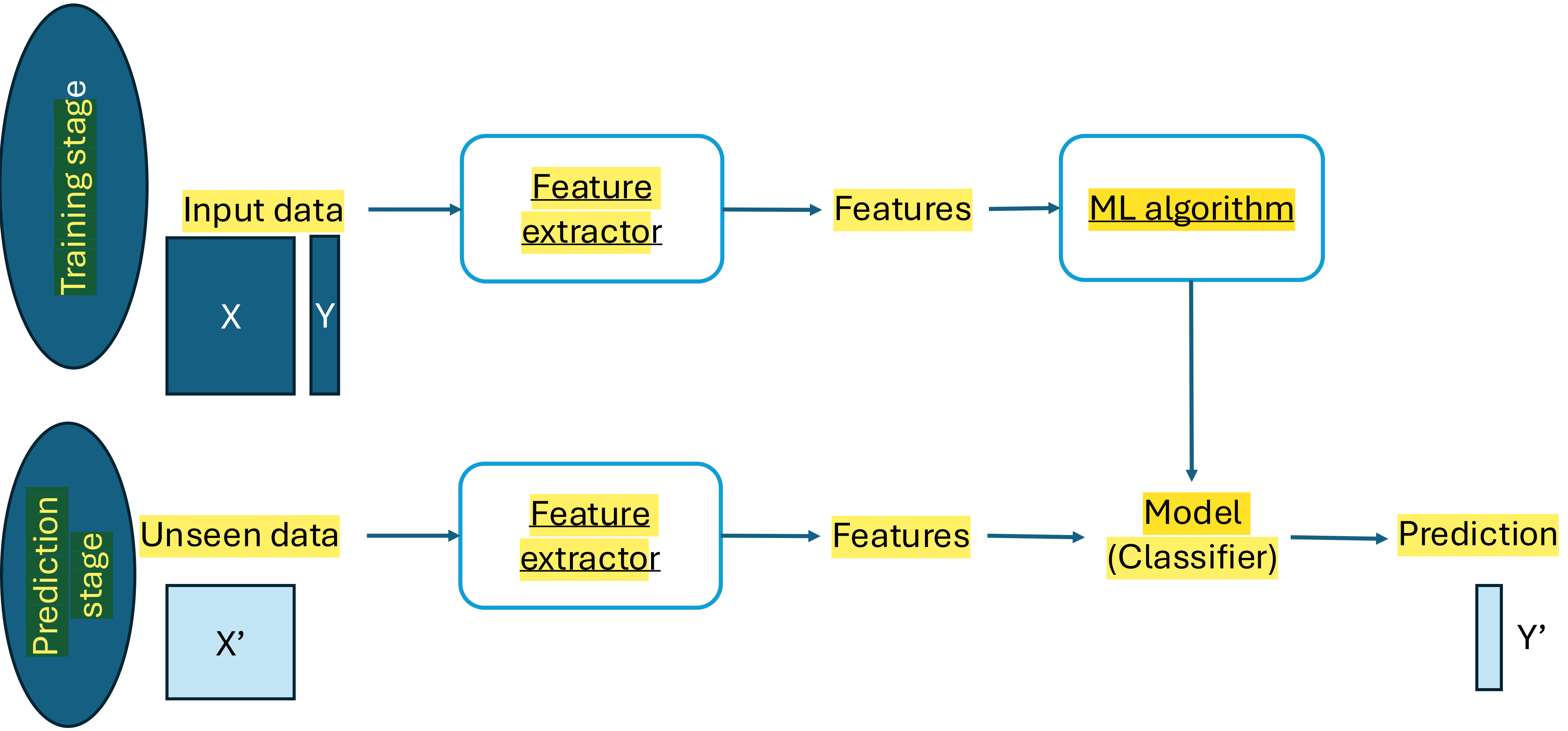
ML workflow


Typically ’prediction’ type
‘Big enough’ data
Evaluation metrics
A problem to solve
Collect & process **data**
Code learning/training **method**
Train, evaluate, and tune **model**
Deploy model
Supervised, un-supervised, semi-supervised, SSL, RL

ML workflow (model-centric AI)

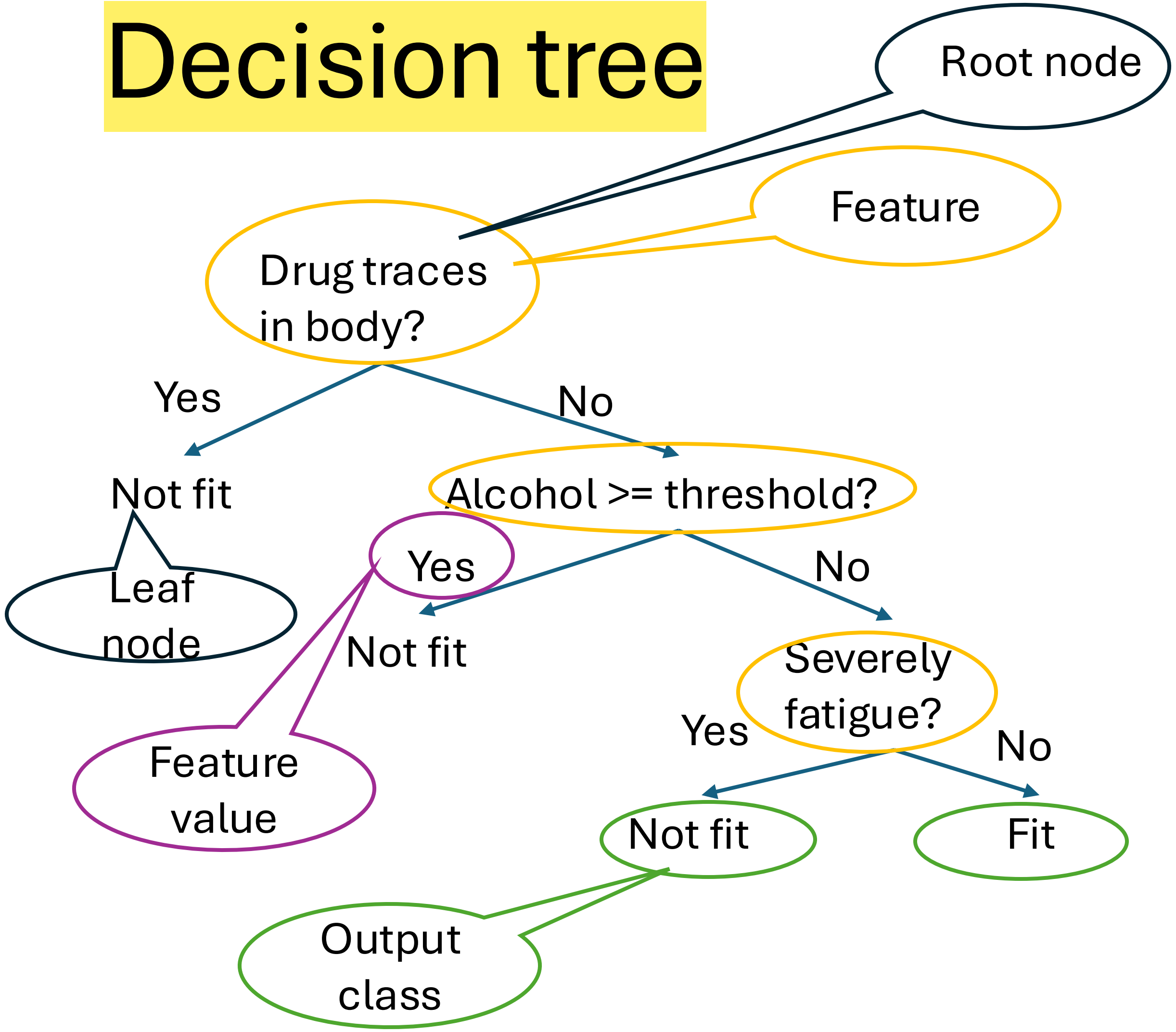


ML workflow (for supervised learning)



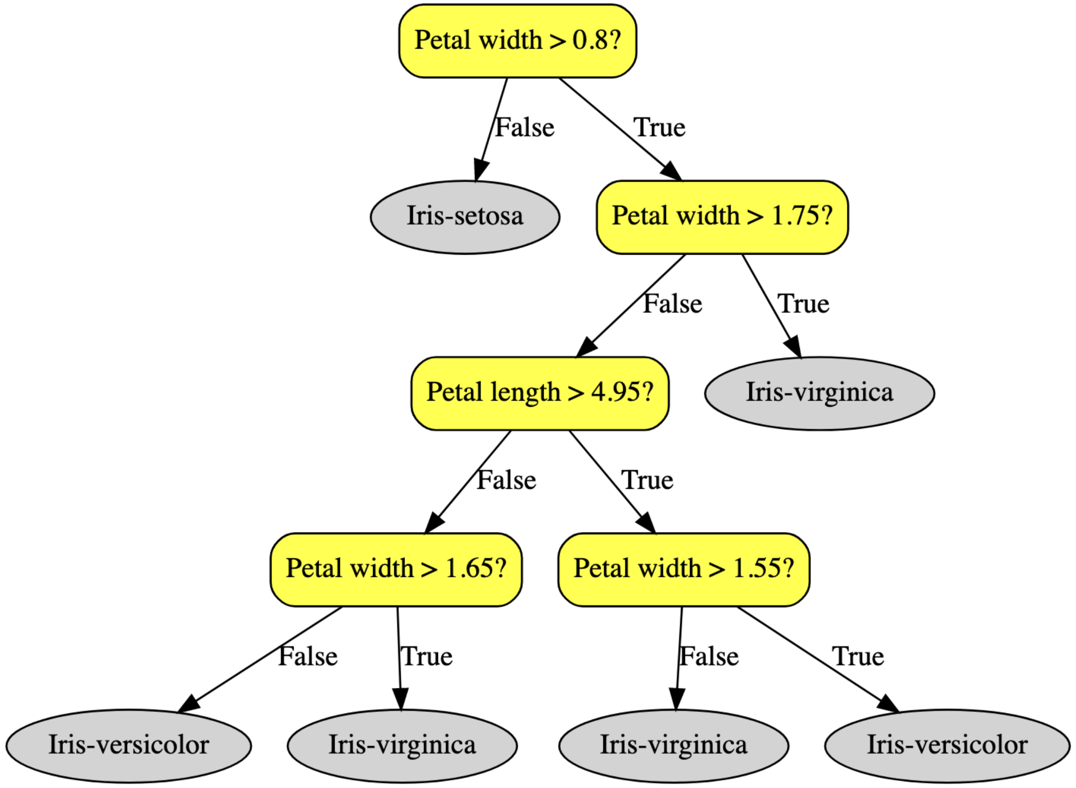
Decision tree

Decision tree



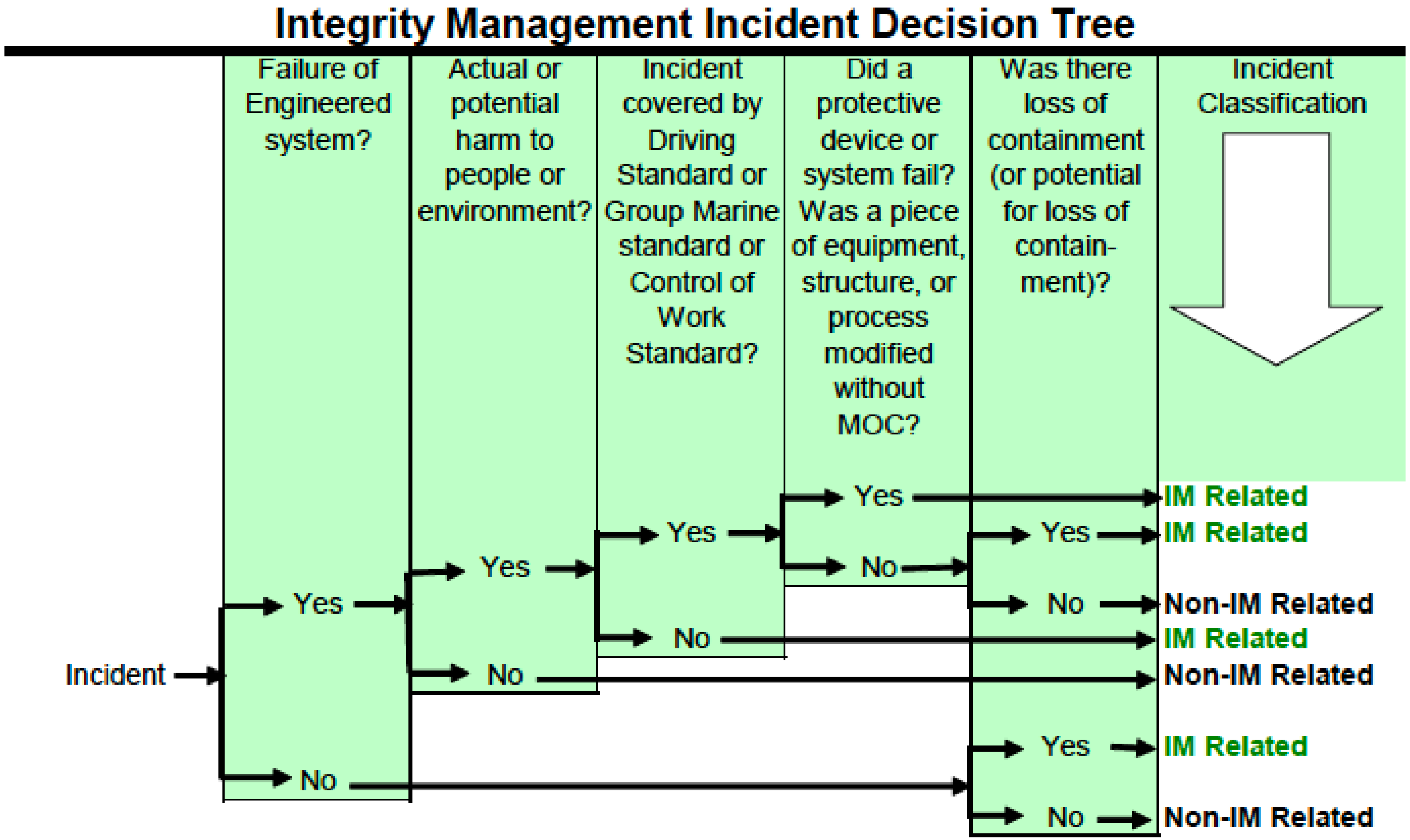
- What is decision tree?
  - Nested if … then statements
- Why use tree?
  - Simple to explain
  - White-box approach

Decision tree



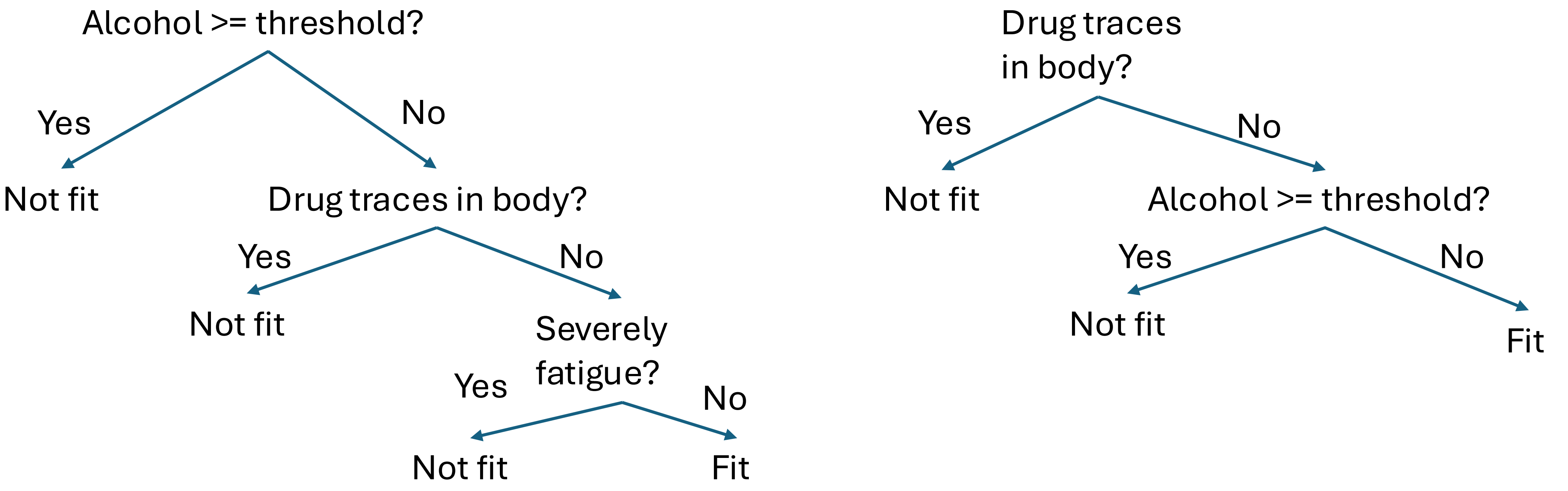
- Where is it used?
  - Diagnosis of patients' disease
  - Deciding to give loan or not
  - Deciding if drivers are fit/not fit to drive

Another example

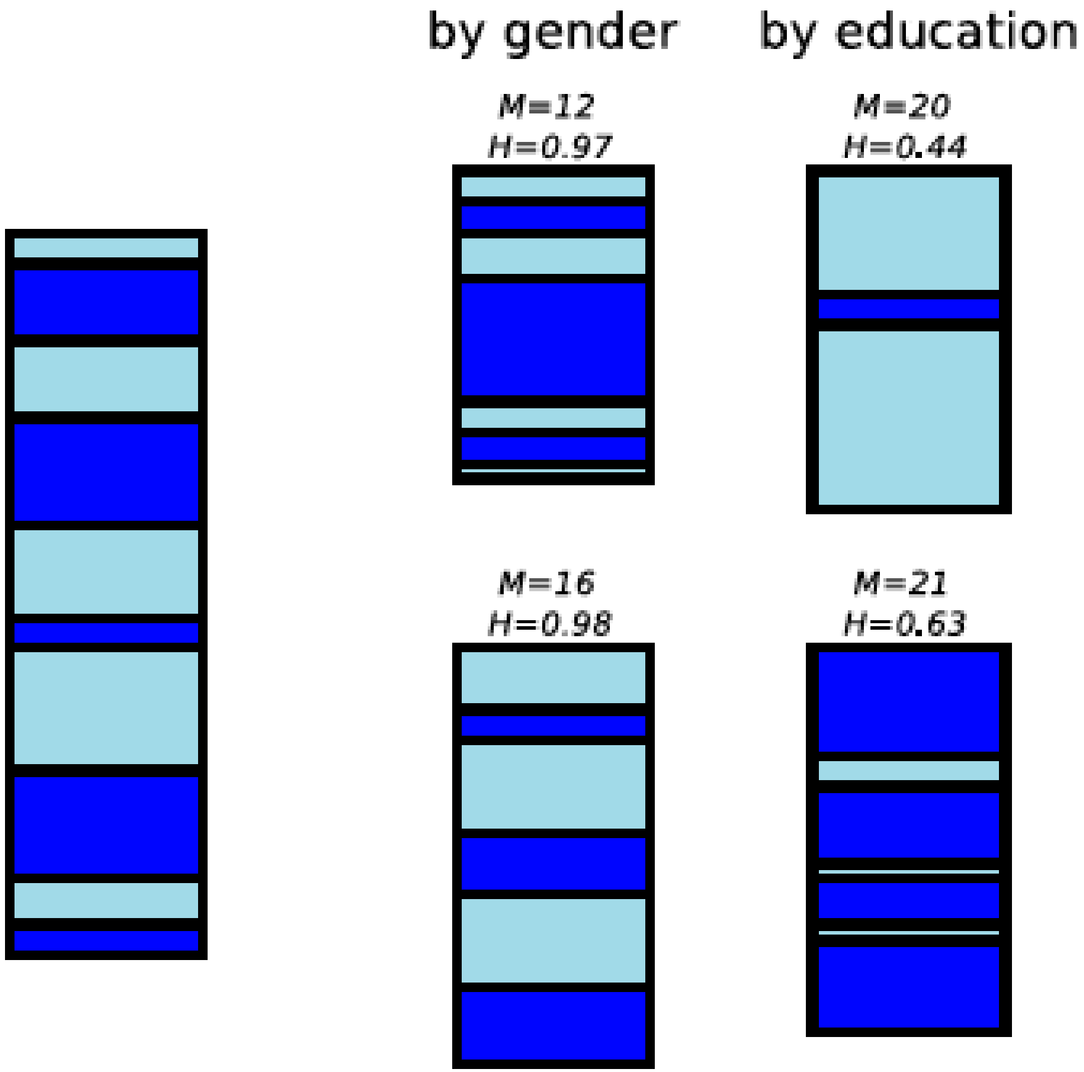


BP U.S. Pipelines and Logistics (USPL) Incident Reporting Safety Manual, 2017

How to grow a decision tree?



- Start at the top of the tree.
- Grow it by splitting attributes one by one. Use ”note impurity” to decide which feature to split on.
- Assign leaf nodes the majority vote in the leaf.
- Prune the tree at the end.



Borrowed from example by Richard Johansson

Discussion

- We have a training set that includes three features, and we'd like to predict a binary output variable. Here is the whole training set.
- If we train a decision tree classifier using this training set, which feature would be considered at the first "branch" of the decision tree (the top node)? Please explain why.

x1	x2	x3	y
A	X	P	True
A	X	Q	True
A	Y	P	False
A	Y	Q	False
B	X	P	True
B	X	Q	False
B	Y	P	False
B	Y	Q	False

Discussion

- Consider a dataset consisting of categorical features and a decision tree classifier trained on this dataset without pruning, using the standard algorithm.

  - If a feature is statistically independent of the target variable, how does this impact the appearance of the feature in the decision tree? Justify your answer.

  - Suppose one of the features has a unique value in each record. Explain how this might affect the structure of the decision tree.

See Notebook

Next class session

- More on decision tree & its recent development
- Ensemble
- Random forest